



Cambridge International AS & A Level

PHYSICS

9702/42

Paper 4 A Level Structured Questions

May/June 2022

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Annotations

✓	Indicates the point at which a mark has been awarded.
X	Indicates an incorrect answer or a point at which a decision is made not to award a mark.
XP	Indicates a physically incorrect equation ('incorrect physics'). No credit is given for substitution, or subsequent arithmetic, in a physically incorrect equation.

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ECF	Indicates 'error carried forward'. Answers to later numerical questions can always be awarded up to full credit provided they are consistent with earlier incorrect answers. <u>Within</u> a section of a numerical question, ECF can be given after AE, TE and POT errors, but not after XP.
AE	Indicates an arithmetic error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
POT	Indicates a power of ten error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
TE	Indicates incorrect transcription of the correct data from the question, a graph, data sheet or a previous answer. For example, the value of 1.6×10^{-19} has been written down as 6.1×10^{-19} or 1.6×10^{19} . Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
SF	Indicates that the correct answer is seen in the working but the final answer is incorrect as it is expressed to too few significant figures.
BOD	Indicates that a mark is awarded where the candidate provides an answer that is not totally satisfactory, but the examiner feels that sufficient work has been done ('benefit of doubt').
CON	Indicates that a response is contradictory.
I	Indicates parts of a response that have been seen but disregarded as irrelevant.
MO	Indicates where an A category mark has not been awarded due to the M category mark upon which it depends not having previously been awarded.
^	Indicates where more is needed for a mark to be awarded (what is written is not wrong, but not enough). May also be used to annotate a response space that has been left completely blank.
SEEN	Indicates that a page has been seen.

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Question	Answer	Marks
1(a)(i)	work (done) per unit mass	B1
	work (done on mass) in moving mass from infinity (to the point)	B1
1(a)(ii)	$E_P = \phi m$ $E_P = (-GM/r) \times m = -GMm/r$ or $\phi = -GM/r$ and $E_P = \phi m = -GMm/r$	B1
1(b)(i)	$\Delta E_P = 6.67 \times 10^{-11} \times 1.99 \times 10^{30} \times 2.20 \times 10^{14} \times [1/(6.38 \times 10^{10}) - 1/(8.44 \times 10^{11})]$	C1
	$= 4.23 \times 10^{23} \text{ J}$	A1
1(b)(ii)	(gravitational) force is attractive so decrease or (gravitational) force does work so decrease	B1
1(b)(iii)	$\Delta E_P = \frac{1}{2}m(v_2^2 - v_1^2)$	C1
	$4.23 \times 10^{23} = \frac{1}{2} \times 2.20 \times 10^{14} \times (v^2 - 34\,100^2)$	C1
	$v (= 70\,800 \text{ m s}^{-1}) = 70.8 \text{ km s}^{-1}$	A1
1(c)	both PE and KE equations include m , so path is unchanged	B1

Question	Answer	Marks
2(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
2(b)(i)	(electric) force is perpendicular to velocity (of particles)	B1
	force (perpendicular to velocity) causes centripetal acceleration or force does not change the speed of the particles or force has constant magnitude	B1
2(b)(ii)	$F = e^2 / 4\pi\epsilon_0 x^2$	C1
	$= (1.60 \times 10^{-19})^2 / [4\pi \times 8.85 \times 10^{-12} \times (2 \times 1.59 \times 10^{-10})^2]$	A1
	$= 2.28 \times 10^{-9} \text{ N}$	
2(b)(iii)	$F = m\omega^2$ and $\omega = 2\pi / T$ or $F = mv^2 / r$ and $v = 2\pi r / T$	C1
	$F = 4\pi^2 mr / T^2$	C1
	$T = \sqrt{[4\pi^2 \times 9.11 \times 10^{-31} \times 1.59 \times 10^{-10} / (2.28 \times 10^{-9})]}$	
	$= 1.58 \times 10^{-15} \text{ s}$	A1
2(c)(i)	- electron and positron interact - positron is anti-particle of electron (pair) annihilation occurs <i>Any two points, 1 mark each</i>	B2
	mass of the electron and positron converted into photon energy	B1
2(c)(ii)	PET scanning	B1

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Question	Answer	Marks
3(a)	(thermal) energy per unit mass	B1
	energy to change state between liquid and gas at constant temperature	B1
3(b)(i)	$q = mL = 0.37 \times 2.3 \times 10^6$ $= 8.5 \times 10^5 \text{ J}$	A1
3(b)(ii)	$pV = nRT$ and $T = 373 \text{ K}$	C1
	$n = 370 / 18$	C1
	$V = [(370 / 18) \times 8.31 \times 373] / (1.0 \times 10^5) = 0.64 \text{ m}^3$	A1
3(b)(iii)	$w = p\Delta V$	C1
	$= 1.0 \times 10^5 \times 0.64$	A1
	$= 6.4 \times 10^4 \text{ J}$	
3(b)(iv)	(water does work against atmosphere so) work done on water is negative	B1
	increase in internal energy $= (8.5 - 0.64) \times 10^5 = 7.9 \times 10^5 \text{ J}$	A1
3(c)	valid reasoning of how work done by water is affected	M1
	correct use of first law to draw conclusion about effect on specific latent heat that is consistent with work done	A1

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Question	Answer	Marks
4(a)	oscillations (of object) at maximum amplitude	B1
	when driving frequency equals natural frequency (of object)	B1
4(b)(i)	$T = 2\pi / \omega$	C1
	$= 2\pi / 5.0\pi$	A1
	$= 0.40 \text{ s}$	
4(b)(ii)	displacement scale labelled $-1.0, -0.5, (0), 0.5, 1.0$ on the 2 cm tick marks	B1
	t scale labelled $0.2, 0.4, 0.6, 0.8, 1.0, 1.2$ on the 2 cm tick marks	B1
4(b)(iii)	$\phi = 2\pi\Delta t / T$	C1
	$= 2\pi \times 0.10 / 0.40$ or $2\pi \times 0.30 / 0.40$	A1
	$= 1.6 \text{ rad}$ or 4.7 rad	

Question	Answer	Marks
5(a)	charge / potential (difference)	M1
	charge is charge on one plate, <u>and</u> potential is p.d. across the plates	A1
5(b)	p.d. across both capacitors = E	B1
	$Q_T = Q_1 + Q_2$	B1
	$C_T E = C_1 E + C_2 E$ hence $C_T = C_1 + C_2$	B1
5(c)(i)	$[(1/22) + (1/47)]^{-1} = 15 \mu\text{F}$	A1
5(c)(ii)	energy = $\frac{1}{2}CV^2$	C1
	$= \frac{1}{2} \times 15 \times 10^{-6} \times 12^2$	A1
	$= 1.1 \times 10^{-3} \text{ J}$	
5(c)(iii)	initial p.d. (across $22 \mu\text{F}$) = $12 \times (15/22)$ $= 8.2 \text{ V}$ or final p.d. across both capacitors = $6.0 \times (22/15)$ $= 8.8 \text{ V}$	C1
	$V = V_0 \exp[-t/(2.7 \times 10^6 \times 15 \times 10^{-6})]$	C1
	$6.0 = 8.2 \exp[-t/(2.7 \times 10^6 \times 15 \times 10^{-6})]$ or $8.8 = 12 \exp[-t/(2.7 \times 10^6 \times 15 \times 10^{-6})]$	A1
	$t = 13 \text{ s}$	

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Question	Answer	Marks
6(a)	there must be a current (in the wire)	B1
	(wire) must be at a non-zero angle to the magnetic field	B1
6(b)(i)	arrow from X pointing horizontally to the left	B1
	arrow from Y pointing diagonally upwards and to the left at about 45°	B1
	arrow from Z pointing horizontally to the right	B1
6(b)(ii)	(flux densities at W and X are approximately) equal	B1
	(flux density at) Y greater than (flux density at) Z	B1
6(c)	current in wire creates magnetic field around wire	B1
	(each) wire sits in the magnetic field created by the other	B1
	(for each wire,) current / wire is perpendicular to magnetic field (due to other wire), (so) experiences a (magnetic) force	B1

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Question	Answer	Marks
7(a)	induced e.m.f. is (directly) proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
7(b)	V_2 stepped, all at non-zero values, between $t = 0$ and $t = 0.40$ s	B1
	V_2 shown with same non-zero magnitude up to $t = 0.15$ s and after $t = 0.25$ s but with a different magnitude between these times	B1
	V_2 shown with a magnitude between $t = 0.15$ s and $t = 0.25$ s that is three times the magnitude before $t = 0.15$ s and after $t = 0.25$ s	B1
	V_2 shown with same sign up to $t = 0.15$ s and after $t = 0.25$ s, and opposite sign in between	B1
7(c)(i)	changing current in coil causes changing (magnetic) field or changing (magnetic) flux causes induced e.m.f. in ring	B1
	induced e.m.f. in ring causes current in ring	B1
	(magnetic) field due to (induced) current in ring interacts with (coil's) field to cause upwards force (on ring) or (induced) current in ring perpendicular to (coil's magnetic) field causes upwards force (on ring)	B1
7(c)(ii)	both magnetic fields reverse direction so ring still jumps up or current (in ring) and (coil's) field both reverse so ring still jumps up	B1

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Question	Answer	Marks
8(a)(i)	photoelectric effect	B1
8(a)(ii)	electron diffraction	B1
8(b)(i)	$\lambda = h / p$	C1
	$p = 4 \times 1.66 \times 10^{-27} \times 6.2 \times 10^7$ (= $4.1 \times 10^{-19} \text{ N s}$)	C1
	$\lambda = 6.63 \times 10^{-34} / 4.1 \times 10^{-19}$ = $1.6 \times 10^{-15} \text{ m}$	A1
8(b)(ii)	line with negative gradient throughout	B1
	curve asymptotic to both axes with non-zero λ at $v = 6.2 \times 10^7 \text{ m s}^{-1}$	B1
8(c)	(de Broglie) wavelength negligible compared with width of doorway	B1

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Question	Answer	Marks
9(a)(i)	speed is (directly) proportional to distance	M1
	where speed is speed of recession of galaxy (from observer) and distance is distance of galaxy away from observer	A1
9(a)(ii)	wavelengths (of spectral lines) are greater (than their known values)	B1
	redshift shows stars (in distant galaxies) moving away from Earth	B1
9(b)	(all) parts of Universe moving away from each other	B1
	more distant objects are moving away faster	B1
	matter must have been close together / very dense in the past	B1

Question	Answer	Marks
10(a)	spontaneous emission of (ionising) radiation	B1
	emission from unstable nucleus	B1
10(b)(i)	curve with decreasing negative gradient passing through $(0, N_0)$	B1
	curve passing through $(T, 0.5N_0)$	B1
	curve passing through $(2T, 0.25N_0)$ and $(3T, 0.125N_0)$	B1
10(b)(ii)	line through origin with positive gradient	B1
	straight line passing through (N_0, A_0)	B1
10(c)(i)	activity	B1
10(c)(ii)	decay constant	B1
10(d)	$N = N_0 \exp(-\ln 2 \times 1.70T / T)$	C1
	$N / N_0 = 0.31$	A1